

Assignment NO 8:

Write PL/SQL Trigger for the creation of insert trigger, delete trigger and update trigger on the given problem statements.

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Panel – C [Batch C3]

Problem Statement:

Q.1) Consider the following relational schema:

- **parent(id int primary key, refCount int);**
- **child(id int primary key, parent_id int);**

Write a trigger that updates the refcount whenever we insert or delete into the child table.

```
mysql> create database exp8;
```

```
Query OK, 1 row affected (0.04 sec)
```

```
mysql> use exp8;
```

```
Database changed
```

```
mysql> create table parent (
```

```
-> id int primary key,
```

```
-> refcount int default 0
```

->);

Query OK, 0 rows affected (0.06 sec)

mysql> create table child (

-> id int primary key,

-> parent_id int,

-> foreign key (parent_id) references parent(id)

->);

Query OK, 0 rows affected (0.06 sec)

mysql> insert into parent values (1,0),(2,0),(3,0);

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from parent;

```
+----+-----+
| id | refcount |
+----+-----+
| 1 | 0 |
| 2 | 0 |
| 3 | 0 |
+----+-----+
```

3 rows in set (0.00 sec)

mysql> delimiter \$\$

```
mysql> create trigger child_after_insert
```

```
-> after insert on child
```

```
-> for each row
```

```
-> begin
```

```
->   update parent
```

```
->   set refcount = refcount + 1
```

```
->   where id = new.parent_id;
```

```
-> end$$
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> delimiter ;
```

```
mysql> delimiter $$
```

```
mysql> create trigger child_after_delete
```

```
-> after delete on child
```

```
-> for each row
```

```
-> begin
```

```
->   update parent
```

```
->   set refcount = refcount - 1
```

```
->   where id = old.parent_id;
```

```
-> end$$
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> delimiter ;
```

```
mysql> insert into child values (101,1);
```

```
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into child values (102,1);
```

Query OK, 1 row affected (0.01 sec)

mysql> insert into child values (103,2);

Query OK, 1 row affected (0.01 sec)

mysql> select * from parent;

id	refcount
1	2
2	1
3	0

3 rows in set (0.00 sec)

mysql> delete from child where id = 101;

Query OK, 1 row affected (0.01 sec)

mysql> select * from parent;

id	refcount
1	1
2	1
3	0

3 rows in set (0.00 sec)

```
Command Prompt - mysql -u root -p
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CHIMU>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database exp8;
Query OK, 1 row affected (0.04 sec)

mysql> use exp8;
Database changed
mysql> create table parent (
  ->   id int primary key,
  ->   refcount int default 0
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> create table child (
  ->   id int primary key,
  ->   parent_id int,
  ->   foreign key (parent_id) references parent(id)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> insert into parent values (1,0),(2,0),(3,0);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from parent;
+-----+
| id | refcount |
+-----+
| 1 | 0 |
+-----+
```

```
mysql> select * from parent;
+-----+
| id | refcount |
+-----+
| 1 | 0 |
| 2 | 0 |
| 3 | 0 |
+-----+
3 rows in set (0.00 sec)

mysql> delimiter $$
mysql> create trigger child_after_insert
  -> after insert on child
  -> for each row
  -> begin
  ->   update parent
  ->   set refcount = refcount + 1
  ->   where id = new.parent_id;
  -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> delimiter ;
mysql> delimiter $$
mysql> create trigger child_after_delete
  -> after delete on child
  -> for each row
  -> begin
  ->   update parent
  ->   set refcount = refcount - 1
  ->   where id = old.parent_id;
  -> end$$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> delimiter ;
mysql> insert into child values (101,1);
Query OK, 1 row affected (0.02 sec)
```


Q.2) Consider the following relational schema:

- **Project(Proj_id, Proj_name, Proj_duration, Proj_status)**
- **archived_projects(Proj_id, Proj_Time, Proj_Status)**

Write a trigger for

When a project in a projects table is marked as completed, automatically move its details to an archived_projects table for historical records.

```
mysql> create table project (  
->  proj_id int primary key,  
->  proj_name varchar(50),  
->  proj_duration int,  
->  proj_status varchar(20)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> create table archived_projects (  
->  proj_id int,  
->  proj_time int,  
->  proj_status varchar(20)  
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> insert into project values  
-> (1,'ai system',6,'ongoing'),  
-> (2,'web app',4,'ongoing'),  
-> (3,'mobile app',5,'ongoing');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> delimiter $$
```

```
mysql> create trigger project_after_update
```

```
-> after update on project
```

```
-> for each row
```

```
-> begin
```

```
-> if new.proj_status = 'completed' and old.proj_status <> 'completed'
then
```

```
-> insert into archived_projects(proj_id, proj_time, proj_status)
```

```
-> values (new.proj_id, new.proj_duration, new.proj_status);
```

```
-> end if;
```

```
-> end$$
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> delimiter ;
```

```
mysql> update project
```

```
-> set proj_status = 'completed'
```

```
-> where proj_id = 1;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from archived_projects;
```

```
+-----+-----+-----+
```

```
| proj_id | proj_time | proj_status |
```

```
+-----+-----+-----+
```

```
| 1 | 6 | completed |
```

```
+-----+-----+-----+
```


1 row in set (0.00 sec)

```
mysql> select * from project;
```

proj_id	proj_name	proj_duration	proj_status
1	ai system	6	completed
2	web app	4	ongoing
3	mobile app	5	ongoing

3 rows in set (0.00 sec)

```
mysql> drop trigger project_after_update;
```

Query OK, 0 rows affected (0.02 sec)

```
Command Prompt - mysql -u root -p
-----+-----+
2 rows in set (0.02 sec)

mysql> create table project (
  ->   proj_id int primary key,
  ->   proj_name varchar(50),
  ->   proj_duration int,
  ->   proj_status varchar(20)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> create table archived_projects (
  ->   proj_id int,
  ->   proj_time int,
  ->   proj_status varchar(20)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> insert into project values
  -> (1,'ai system',6,'ongoing'),
  -> (2,'web app',4,'ongoing'),
  -> (3,'mobile app',5,'ongoing');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> delimiter $$
mysql> create trigger project_after_update
  -> after update on project
  -> for each row
  -> begin
  ->   if new.proj_status = 'completed' and old.proj_status <> 'completed' then
  ->     insert into archived_projects(proj_id, proj_time, proj_status)
  ->     values (new.proj_id, new.proj_duration, new.proj_status);
  ->   end if;
  -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;
mysql> update project
  -> set proj_status = 'completed'
  -> where proj_id = 1;
Query OK, 1 row affected (0.01 sec)
```

```
Command Prompt - mysql -u root -p

mysql> delimiter $$
mysql> create trigger project_after_update
  -> after update on project
  -> for each row
  -> begin
  ->   if new.proj_status = 'completed' and old.proj_status <> 'completed' then
  ->     insert into archived_projects(proj_id, proj_time, proj_status)
  ->     values (new.proj_id, new.proj_duration, new.proj_status);
  ->   end if;
  -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;
mysql> update project
  -> set proj_status = 'completed'
  -> where proj_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from archived_projects;
+-----+-----+
| proj_id | proj_time | proj_status |
+-----+-----+
| 1       | 6         | completed   |
+-----+-----+
1 row in set (0.00 sec)

mysql> drop trigger project_after_update;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from project;
+-----+-----+-----+-----+
| proj_id | proj_name | proj_duration | proj_status |
+-----+-----+-----+-----+
| 1       | ai system | 6             | completed   |
| 2       | web app   | 4             | ongoing     |
| 3       | mobile app | 5             | ongoing     |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```